

Heterogeneous Dual-Dynamic Attention Network for Modeling Mutual Interplay of Stocks

Haotian Liu[✉], Yadong Zhou[✉], *Member, IEEE*, Yuxun Zhou[✉],
and Bowen Hu[✉], *Graduate Student Member, IEEE*

Abstract—Modern quantitative finance and portfolio-based investment hinge on the dependence structure among financial instruments (like stocks) for return prediction, risk management, and hedging. Several attempts have been made to leverage recent advancements in graph neural networks (GNNs) to achieve effective modeling to capture stock interactions. However, existing work suffers from the homogeneity and stationary of the predefined graph structure, and their reliance on hard-coded correlation-based relationships neglects the mutual intervention of different instruments. In this study, we propose the heterogeneous dual-dynamic attention network (HD²AT) to address these challenges in modeling the mutual interplay among stocks, across four key dimensions: heterogeneity, causality, structural dynamics, and lead-lag dynamics. More specifically, our method uncovers the causal interdependencies among instruments, dynamically inferring the heterogeneous network structure from real-time trading activities. Subsequently, we identify the cross-temporal propagation of interventions by integrating interactions at the node, time, and graph levels. Extensive experiments on real-world trading data demonstrate the superiority of the proposed framework over various baselines in terms of both return prediction and back-testing profitability.

Impact Statement—Stock return prediction holds significant importance for investors and governments, as it offers opportunities to increase profits and provides insights into future economic conditions. However, the dynamic complexity of the stock market poses challenges in developing effective solutions. This article proposes a method that models the causal relationships driving mutual interactions among stocks, rather than superficial correlations. Furthermore, we combine structural dynamics and temporal dynamics to model the propagation of these interactions within stock networks. By adapting to the ever-volatile stock market, our method enhances stock return prediction accuracy while mitigating investment risks. As such, the prediction results serve as valuable references for diverse stakeholders.

Index Terms—Dynamic attention, graph neural networks (GNNs), graph structure inference, stock return prediction.

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The authors are with the Ministry of Education Key Laboratory for Intelligent Networks and Network Security, Faculty of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: yadongzhou@gmail.com).

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I. INTRODUCTION

THE financial market, particularly the stock market, is a real-world multiagent, multimodal, and interconnected system involving numerous automatic or manual transactions among investors, speculators, entity owners, etc. Stock return prediction has long piqued the curiosity of researchers, as investors persistently seek to anticipate the future trends of listed companies in their quest for maximizing profits within the financial market [1], [2], [3], [4]. Prior studies demonstrated the power of sequential models on stock prediction [5], [6]. Nonetheless, as the stock return is intrinsically nonstationary and highly volatile, delineating the complex interactions behind stock price comovement remains challenging in both quantitative finance and statistical machine learning.

Typically, each stock is treated as a sequence of time-series prices (e.g., opening price, closing price, volume, etc). In this scenario, technical analysis techniques relying on recurrent neural networks (RNNs) were widely employed to learn the stock representations [5], [7], [8], [9], [10]. These approaches assume that each stock is mutually exclusive, ignoring that each instrument is inevitably affected by related entities due to factor movement, lead-lag effect, portfolio rebalancing, etc. [5], [11], [12], [13]. Since the advent of graph neural networks (GNNs), recent research on stock prediction proved the effectiveness of graph-based methods [14], [15], [16], where each instrument is modeled as a node and the edge between two nodes is used to represent their relations/interactions. However, most stock relationships are directly modeled by correlation, such as price correlation [16], companies relations [11], [17], and news cooccurrence [4]. They rely on measuring statistical associations rather than delving into the authentic mechanisms that drive these relationships, thus providing only a superficial understanding of the complicated interventions among stocks. Moreover, the prevalent practice of employing predefined hard-coded graphs in these methodologies falls short of capturing the intricate temporal dynamics and evolving structures inherent in price-volume trading data. This limitation not only hinders the model's ability to comprehend the complex web of interactions within the stock market but also causes it to overlook the heterogeneous, multihop, and dynamic propagation of time-level interactions among different stocks.

In this study, we aim to dynamically model the heterogeneous mutual interplay of stocks and seamlessly integrate node-, time-, and graph-level representations for stock return

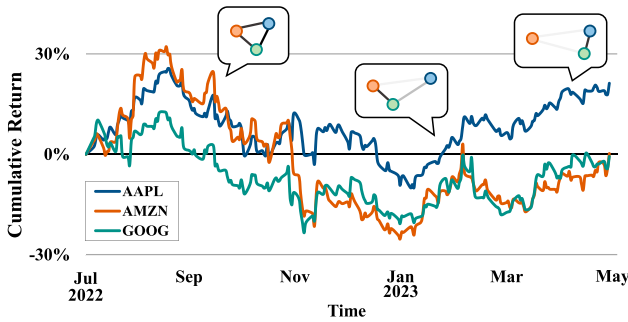


Fig. 1. Illustration of the connection strength among different stocks varies with time, where the linkages are led by market status.

prediction. However, this task is nontrivial, presenting us with three challenges. 1) *Modeling dynamic stock networks*. Static graphs fail to account for real-time market dynamics, including phenomena like factor events. As depicted in Fig. 1, the interfirm linkages can change rapidly over time due to daily trading activities. For instance, the graph started with a dense structure among Apple (AAPL), Amazon (AMZN), and Google (GOOG), but only two of them persisted with a significant connection toward the end of the period. This highlights a critical limitation of using hard-coded graphs, as they lack the adaptability to time-varying stock networks. Therefore, these static graphs cannot harness the full spectrum of cross-sectional information available in dynamic financial markets. 2) *Modeling heterogeneous stock interventions*. Predefined correlation-based relationships cannot reflect the real interplay among stocks, which often manifests as interventions in various forms like price and volume momentum spillovers. Consequently, correlation graphs would be incomplete and noisy, if not impossible, to model real-world stock interactions. 3) *Modeling time-aware impact of heterogeneous interventions*. An individual stock not only reacts to its historical signals but also exhibits a certain level of price offset due to interventions it has received in the preceding days, owing to the lead-lag effect. However, when analyzing these dynamics, we can only observe the combined outcomes of both factors. This presents a challenge when modeling the cross-temporal propagation of interventions.

To tackle these challenges, we propose the heterogeneous dual-dynamic attention network (HD²AT), a market-driven framework for stock return prediction and quantitative trading¹. In HD²AT, intervention graphs are dynamically inferred from the market signals in the current transaction period. This network structure portrays a multitude of heterogeneous causal interactions among stocks, enabling the dynamic modeling of the intricate intervention propagation within the market. Based on these heterogeneous graphs, HD²AT models the responses generated by stocks' own technical indicators and the offsets resulting from the interventions they receive separately. By leveraging time-aware dynamic attention and graph-aware heterogeneous attention, our framework achieves accurate intervention propagation and aggregation at the node level, time level, and graph level. In summary, HD²AT introduces a

“dual-dynamic” perspective within financial markets by modeling both the spatial structural dynamics and the temporal lead-lag dynamics. Furthermore, our methodology can be extended to diverse multivariate time-series analysis and time-varying graph learning domains beyond finance, such as neurological connectivity network estimation [18], [19], topology tracking for power systems [20], [21], and latent social network inference [22], [23].

We would like to emphasize three primary contributions as follows.

- 1) To model the varying market topology, we dynamically infer the multiday stock network by the real-time market signals. These graphs not only integrate temporal information on the evolution of the stock network but also encompass intricate heterogeneous structures within the market.
- 2) To infuse mutual interplay in the connection among stocks, a series of intervention estimators are designed to mine causal effects rather than correlations. This modeling not only preserves directional information but is also more reliable and statistically robust for identifying the intervention-driven offset component.
- 3) We decouple market signals into responses and offsets to precisely portray the impact of each intervention. Equipped with the proposed time-aware and graph-aware attention mechanisms, HD²AT implements dynamic heterogeneous attention from both posterior and prior perspectives.

II. RELATED WORK

A. Time Series Stock Prediction

In stock price prediction, traditional techniques often prioritize time-series features, referred to as technical analysis. These approaches involve learning sequential latent representations and applying them to downstream prediction tasks, e.g., stock movement prediction and return prediction. RNNs, such as long short-term memory (LSTM) [24] and gated recurrent unit (GRU) [25], have found extensive use in capturing and representing sequential features. A prevalent strategy is to gather multiple price signals or construct various indicators as inputs, and then apply RNNs to extract technical features for predicting the price of individual stocks [5], [7], [11], [13], [16]. Some studies have also explored the fusion of diverse types of market indicators and fundamental indicators in an endeavor to enhance model performance [2], [11], [26], [27], [28]. For instance, approaches like Num hierarchical transformer model (NumHTML) [29] and multi-view fusion network (MFN) [3] aggregate multimodal features such as technical indicators and textual news to glean investment insights. However, an underlying assumption in these studies is that market signals for all stocks are mutually exclusive. This presumption does not align with the reality of highly interconnected financial markets due to phenomena like momentum spillovers [12].

B. Graph-Based Stock Prediction

Since all entities operate within the same financial system, the price trajectory of each stock is intricately connected with

¹The code of HD²AT can be accessed via the following Github link: <https://github.com/lauht/HD2AT>.

TABLE I
COMPARISON BETWEEN GRAPH-BASED SOTA MODELS AND THE
PROPOSED MODEL

Model	Hetero- geneity	Structural Dynamics	Lead-Lag Dynamics	Causal Interplay
MAC [14]	✗	✗	✗	✗
TGC [17]	✗	✓	✗	✗
HyperStockGAT [35]	✓	✗	✗	✗
MAN-SF [11]	✗	✗	✓	✗
STHAN-SR [34]	✗	✗	✓	✗
ESTIMATE [15]	✓	✗	✓	✗
THGNN [16]	✓	✓	✗	✗
AD-GAT [5]	✗	✓	✓	✗
DANSMP [13]	✓	✓	✗	✗
HD ² AT (ours)	✓	✓	✓	✓

its peer stocks [30]. Typically, this interconnectedness is represented as a stock network, guided by various factors such as price correlation [16], companies relations [11], [17], and news cooccurrence [4]. Subsequently, variants of GNNs are employed to aggregate the influences from related stocks [14], [16], [31], [32], [33]. An obvious limitation of these methods is that they rely on graphs with static structures throughout the entire pipeline. For instance, MAC [14] leverages the graph convolutional network (GCN) to aggregate the effects of news on related companies. ESTIMATE [15], utilizing hypergraphs based on industry classifications, captures nonpairwise correlations among stocks. DANSMP [13] constructs a heterogeneous market knowledge graph and learns interclass and intraclass interactions among listed companies and associated executives. In essence, these techniques depict stock interactions as hard-coded graphs, which lack the dynamic adaptability necessary for trading signals and inhibit the capability to amalgamate cross-sectional information. To overcome these limitations, several efforts have been made to construct graphs that are tailored to specific downstream prediction tasks [34]. One pioneering study on the dynamic relationships among entities is attribute-driven graph attention network (AD-GAT) [5], which adjusts the predefined homogeneous linkage considering the momentum spillover effect. Temporal and heterogeneous graph neural network (THGNN) [16] constructs a temporal and heterogeneous graph comprising multiple relationships between companies based on historical prices. However, these relationships directly modeled by correlations may not accurately reflect the true interactions among stocks.

Table I summarizes the primary strengths of our model, comparing it with various state-of-the-art (SOTA) graph-based models. This comparison not only encompasses spatial aspects relating to heterogeneity, structural dynamics, and the causal interplay within the graph but also considers the temporal modeling of lead-lag dynamics. To our best knowledge, we make the first attempt to comprehensively model the intrinsic interactions among stocks from these four dimensions.

III. PRELIMINARIES

In this section, we introduce the technical indicators as essential stock features for downstream tasks, provide a formal definition of the stock return prediction problem, and discuss the mutual interplay among stocks.

A. Technical Indicators

Transaction data serves as the primary representation of a firm's inherent value and the expectations of investors. Each indicator has demonstrated a certain level of predictive significance for assessing a stock's movement and market sentiment [5]. In this study, the basic indicators include the highest and lowest price, opening and closing price, trade volume, and rankings of these values over 5 days, 20 days, and 60 days. As our framework is designed for stock return prediction rather than price prediction, the relative volatility of prices is deemed more crucial than absolute prices for each stock. To deal with varying price levels, we standardize all these price indicators for each stock (excluding rank values), which are regarded as temporal features. Additionally, we augment the technical data with other indicators derived from historical transaction data, including moving average indicators, momentum indicators, volatility indicators, and volume indicators.

B. Definition

1) *Stock Return Prediction*: Given a set $\mathcal{V} = \{1, \dots, N\}$ of stocks, the return y_i^t of stock $i \in \mathcal{V}$ refers to the profit or loss an investor that an investor experiences on the day t from holding that stock over one trading day. It is typically expressed as a percentage and is calculated based on the variation in the stock's price, represented as follows:

$$y_i^t = \frac{c_i^t - c_i^{t-1}}{c_i^{t-1}}$$

where c_i^t and c_i^{t-1} correspond to the closing price of stock i on day t and the previous day $t - 1$, respectively.

Given the historical technical indicators $[x_i^{t-1}, x_i^{t-2}, \dots]$ for each stock $i \in \mathcal{V}$ from past trading days, the task of stock return prediction is to forecast the return $\hat{y}_i^t$ for each stock on the designated prediction day, denoted as t .

2) *Mutual Interventions of Stocks*: Behavioral finance suggests that investors can be influenced by psychological factors, leading to irrational decisions and market anomalies [36], [37]. Examples include overreaction and underreaction to news or events. These reactions are often directly reflected in various trading activities, such as a continuous rise or fall of the stock price or volume. Then, these implicit reactions are subsequently transmitted through momentum spillover effects, spanning across both time and space to influence related stocks. Therefore, when a stock experiences high trading volume alongside significant positive or negative returns, it may imply different latent events and its impact on other stocks may vary.

In this article, we interpret this transmission as interventions within the framework of causal inference [38]. We define a set of K distinct interventions (or treatments) denoted as $\mathcal{R} = \{1, \dots, K\}$ to model the real-time interactions among stocks. Specifically, we select $K = 12$ interventions, i.e., *high/low trading volume with significant positive/negative return*, *steady/sharp fluctuations in closing prices for continuous 3/6-day*, and *declines/rises in closing prices for continuous 3/6-day*. This intervention inference approach enables us to dynamically quantify the mutual interplay within the stock market

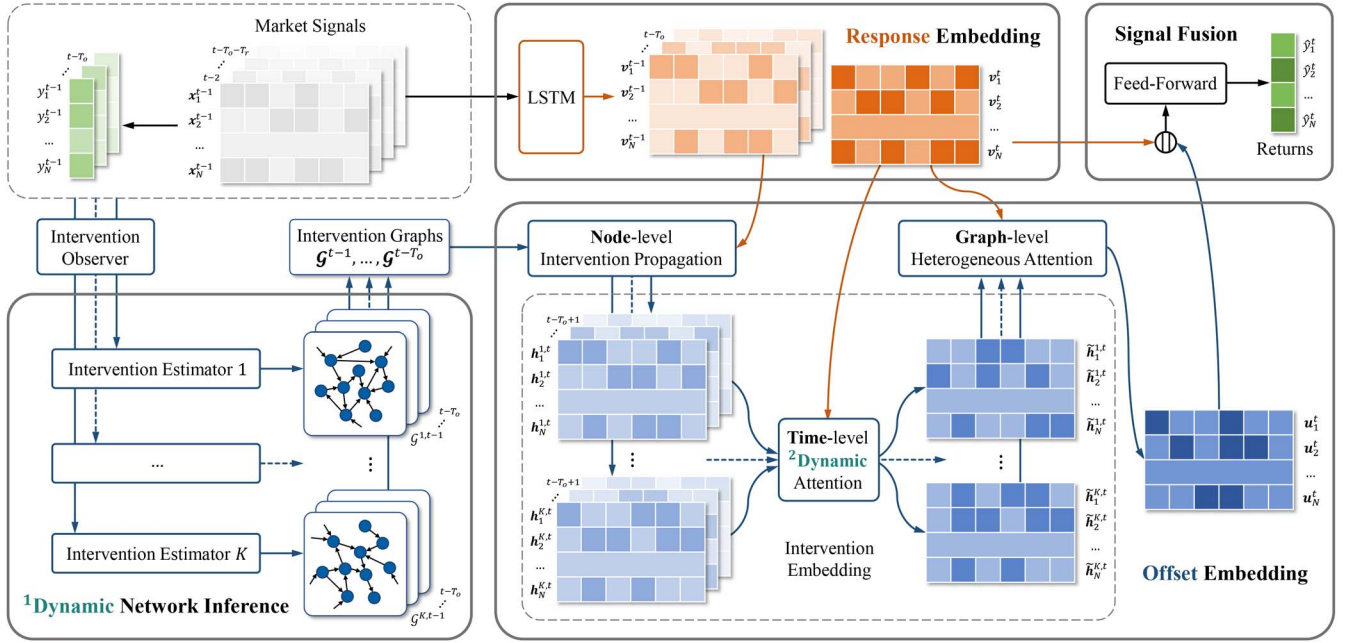


Fig. 2. Pipeline of the proposed HD²AT. First, HD²AT extracts historical return signals from the market signals to dynamically infer heterogeneous intervention graphs through multiple intervention estimators. Then, it derives the historical response sequences for each stock by response embedding. Subsequently, it aggregates intervention information at the node, time, and graph levels by offset embedding. Finally, it combines the response embeddings and offset embeddings to predict stock returns.

and assess how various interventions influence the behavior of stock prices.

a) Discussion on scalability: Indeed, any specific price fluctuation pattern can serve as a candidate for intervention, as long as these patterns empirically or theoretically lead to momentum spillover. However, this pattern needs to accurately reflect the condition of a particular stock, ideally capturing some form of anomaly, as these market anomalies often contain rich potential information. For instance, just as interventions defined by us, a sudden surge in trading volume or a consecutive series of upward movements may likely indicate some underlying events, even potential events with unidentifiable origins. When expanding this methodology, even into multivariate time-series analysis beyond finance, it suffices to define intervention patterns tailored to the specific problem at hand.

3) Intervention Estimators: Given an intervention $k \in \mathcal{R}$, to estimate the mutual intervention effects among stocks, we define the pretrained intervention estimator $\hat{\tau}_{i,j}^k(\cdot) : \mathbb{R}^N \rightarrow \mathbb{R}$, which quantifies the intervention effect of stock j on i for the k th intervention. When the intervention event $w_j^k \in \{0, 1\}$ originating from stock j is observed (with $w_j^k = 1$), the causal effect of this intervention on its peer nodes can be estimated by inputting the returns of all stocks on the trading day $\tilde{x} \in \mathbb{R}^N$ as covariates (confounders). These covariates serve as control variables in the estimation process.

IV. METHODOLOGY

In this section, we provide a detailed introduction to the HD²AT. Fig. 2 offers an overview of the proposed framework for stock return prediction, outlining the four key procedures of our model: 1) *Dynamic network inference*. This

step estimates intervention effects among stocks and creates intervention graphs. 2) *Response embedding*. This step focuses on learning response embeddings that illustrate how stocks respond to historical market signals. 3) *Offset embedding*. This step involves node-, time-, and graph-level aggregations to learn offset embeddings, which quantify the extent of interventions from related stocks. 4) *Signal fusion*. This final step integrates both response and offset embeddings to predict stock returns.

A. Dynamic Network Inference

To infer the dynamic interactions among stocks, we leverage intervention estimators to model the latent intervention, which represents the causal transient-state interdependence.

1) Intervention Estimator: Different from the traditional stock network construction, we leverage metalearners to dynamically generate a heterogeneous graph representing the *de facto* interaction among stocks, rather than correlation. For a target stock i , the intervention effect from source stock j is measured by the T-learner [39]. For a given intervention $k \in \mathcal{R}$, the observation set with L samples can be divided into two distinct groups based on whether the intervention of stock j , w_j^k is observed. These two groups are referred to as the control group, denoted as $\{(\tilde{x}^l, y_i^l)\}_{w_j^k=0}$, and the treated group, denoted as $\{(\tilde{x}^l, y_i^l)\}_{w_j^k=1}$. In both groups, $\tilde{x}^l$ represents the covariate or feature vector, and y_i^l represents the potential outcome for stock i in the l th observation. Note that these K interventions are not mutually exclusive, e.g., a stock can simultaneously experience consecutive price increases and high trading volume. Thus an observation sample may not exclusively belong to a single

treated group. Here, the control response function from stock j to stock i

$$\mu_{i,j}^{k,0}(\tilde{\mathbf{x}}) = \mathbb{E}(y_i | X = \tilde{\mathbf{x}}, w_j^k = 0) \quad (1)$$

is estimated by a base learner, which could be any pretrained supervised regression estimator, e.g., extreme gradient boosting (XGB)Regressor [40]. To alleviate the disturbance from stock i and other stocks, we treat $\tilde{\mathbf{x}} \in \mathbb{R}^N$ as the confounders, which refer to latent variables that affect both intervention and outcome. $\tilde{\mathbf{x}}$ comprises the price-related features of all stocks on the trading day, and the observed outcome y_i is the return of stock i on the following trading day. This setup ensures the intervention has an immediate effect on the return of the target stock. We denote the estimated function as $\hat{\mu}_{i,j}^{k,0}(\cdot)$. Similarly, the intervention response function is denoted as follows:

$$\mu_{i,j}^{k,1}(\tilde{\mathbf{x}}) = \mathbb{E}(y_i | X = \tilde{\mathbf{x}}, w_j^k = 1) \quad (2)$$

which is estimated by a potentially different base learner and the corresponding estimated function is $\hat{\mu}_{i,j}^{k,1}(\cdot)$. Then the causal effect can be imputed from the difference between $\hat{\mu}_{i,j}^{k,1}(\tilde{\mathbf{x}})$ and $\hat{\mu}_{i,j}^{k,0}(\tilde{\mathbf{x}})$, namely $\Delta_{i,j}^k = \hat{\mu}_{i,j}^{k,1}(\tilde{\mathbf{x}}) - \hat{\mu}_{i,j}^{k,0}(\tilde{\mathbf{x}})$. To normalize the effect value and represent it as the weight of a directed edge, the impact of intervention k from stock j to i is mapped by the intervention estimator

$$\hat{\tau}_{i,j}^k(\tilde{\mathbf{x}}) = \begin{cases} 1 - \exp(-\gamma \Delta_{i,j}^k), & \text{if } \Delta_{i,j}^k \geq 0 \\ \exp(\gamma \Delta_{i,j}^k) - 1, & \text{otherwise.} \end{cases} \quad (3)$$

Here, $\hat{\tau}_{i,j}^k(\cdot)$ measures the normalized intervention effect, accounting for the removal of control influence. γ serves as a positive hyperparameter to regulate the linear scaling. Since distinct interventions are linked to specific control response functions, the impact of one intervention is independent of that of others, thereby mitigating information overlap across different estimators.

2) *Dynamic Intervention Graph*: The interaction network of stocks at a given time t can be dynamically represented as a heterogeneous graph $\mathcal{G}^t = (\mathcal{V}, \tilde{\mathcal{E}}^t = \{\mathcal{E}^{k,t}, k \in \mathcal{R}\})$, namely intervention graph, where $\mathcal{V} = \{1, \dots, N\}$ is the node set and $\mathcal{E}^{k,t} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set of intervention k . An intervention subgraph $\mathcal{G}^{k,t} = (\mathcal{V}, \mathcal{E}^{k,t})$ is defined as a subgraph of $\mathcal{G}^t$ that contains all node-relation triples associated with interaction k . To make the interactions comparable over all neighbors in the subgraph, a weighted adjacency matrix of $\mathcal{G}^{k,t-d}$, denoted as $A^{k,t-d} \in \mathbb{R}^{N \times N}$, is constructed. The elements of this matrix are determined as follows:

$$A_{i,j}^{k,t-d} = \begin{cases} \frac{\hat{\tau}_{i,j}^k(\tilde{\mathbf{x}}^{t-d})}{\sum_{n \in \mathcal{V}, w_n^k=1} |\hat{\tau}_{i,n}^k(\tilde{\mathbf{x}}^{t-d})|}, & \text{if } w_j^k = 1 \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Here, $A_{i,j}^{k,t-d}$ represents the normalized effect of intervention k from stock j to i , and $\tilde{\mathbf{x}}^{t-d} = [y_1^{t-d}, \dots, y_N^{t-d}] \in \mathbb{R}^N$ represents the returns of all stocks on the day $t-d$. $\hat{\tau}_{i,j}^k(\cdot)$ is the corresponding intervention estimator, and $w_j^k \in \{0, 1\}$ is a Boolean value indicating whether stock j has an intervention k on other stocks. By utilizing the pretrained intervention estimators, the network inference module can dynamically infer and quantify

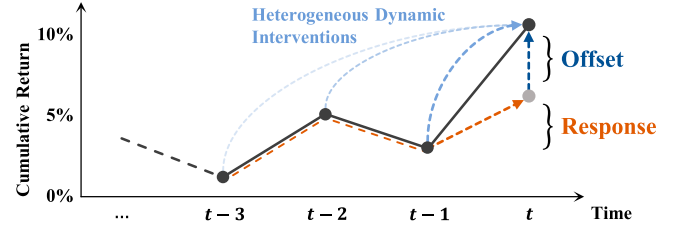


Fig. 3. Visualization of market signals decoupling.

the interactions among stocks based on market signals. This modeling approach allows for the real-time capture of mutual interactions and incorporates heterogeneous interventions, providing a comprehensive source of structural information for downstream graph-based neural networks.

B. Market Signals Decoupling

Stocks not only respond to their own historical signals (technical indicators) but are also subject to momentum spillover effects from related stocks due to market anomalies [26], [28], [34], [35]. In this section, we introduce the concept of decoupling market signals into responses and offsets to accurately portray the impact of each intervention. Fig. 3 provides a visual representation of this market signals decoupling process. The “Response” illustrates the stock’s reaction to its historical signals, while the “Offset” (produced by the heterogeneous dynamic interventions) demonstrates the extent of intervention by neighboring stocks from previous trading days. Existing research has either disregarded offset components or confounded these two types of signals, resulting in models that cannot accurately perceive the underlying dynamics.

1) *Response Embedding*: Technical indicators of each stock are generally represented as a feature vector on the trading day [15], [16], [29]. In this time series modeling, we are to extract temporal relations in an ordered set of continuous price points. Some recent studies mentioned that RNNs could perform better than Transformers [41] in time-series prediction tasks [42]. However, training a separate RNN for each stock within the same framework, particularly with limited sample size in price data, proves to be a formidable task [5]. Following prior research [13], [15], response embedding employs a single LSTM [24] module with two hidden layers to fuse price information in the past T_r trading days to extract transient features from the technical indicators for all stocks. For a given stock $i \in \mathcal{V}$, it fuses the historical signals to generate response embedding $\mathbf{v}_i^t \in \mathbb{R}^{H_v}$, that is

$$\mathbf{v}_i^t = \text{LSTM}(\mathbf{X}_i^{[t-T_r:t]}) \quad (5)$$

where H_v is the hidden size of LSTM. $\mathbf{X}_i^{[t-T_r:t]} = [\mathbf{x}_i^{t-1}, \dots, \mathbf{x}_i^{t-T_r}]$ denotes the matrix concatenating historical signals, where $\mathbf{x}_i^t \in \mathbb{R}^{H_x}$ is the technical indicators of stock i on the t th day. By performing time-series embedding in (5), the learning module learns the response representation of the target stock from a multisequence excitation signal.

2) *Offset Embedding*: To model the effects of intervention within node-, time-, and graph-level propagation, we propose a time-aware heterogeneous attention network to dynamically identify the intervention-driven offset component.

a) *Node-level intervention propagation*: Given a stock $i \in \mathcal{V}$, its received k th intervention at time $t - d$ will affect it at $t - d + 1$, denoted as $\mathbf{h}_i^{k,t-d+1} \in \mathbb{R}^{H_h}$, which is aggregated from neighbors via a linear aggregator

$$\mathbf{h}_i^{k,t-d+1} = \sigma \left(\sum_{n \in \mathcal{V}, n \neq i} A_{i,n}^{k,t-d} \mathbf{W}_{hv}^k \mathbf{v}_n^{t-d} \right) \quad (6)$$

where $A_{i,n}^{k,t-d}$ is the (i, n) element of $\mathbf{A}^{k,t-d}$, $\mathbf{W}_{hv}^k \in \mathbb{R}^{H_h \times H_v}$ is the learned weight matrix, $\mathbf{v}_n^{t-d}$ is the response embedding of stock n and $\sigma(\cdot)$ is a nonlinear activation function. Here, the intervention embedding $\mathbf{h}_i^{k,t-d+1}$ reflects how each stock responds on the next day to its neighboring nodes' intervention. It is worth noting that the intervention graph is established through causal inference, and only the first-order neighboring nodes yield actual causal effects. Hence, this node-level propagation performs only once.

b) *Time-level dynamic attention*: For stocks, it is widely recognized that momentum spillover demonstrates a multihop propagation phenomenon over time, namely the lead-lag effect. Besides, certain types of interventions can also leave a lasting impact on the associated stocks. To aggregate the cumulative cross-time impacts of interventions, we represent the received intervention at time t (the prediction day) with a dynamic attention mechanism as follows:

$$\tilde{\mathbf{h}}_i^{k,t} = \sigma \left(\sum_{d=0}^{T_o-1} \alpha_i^{k,d} \mathbf{W}_{hh}^k \mathbf{h}_i^{k,t-d} \right) \quad (7)$$

where the attention score $\alpha_i^{k,d} \in \mathbb{R}$ indicates the residual influence of the intervention k from day $t - d$ to day t , T_o is the look-back window size for offset embedding and $\mathbf{W}_{hh}^k \in \mathbb{R}^{H_h \times H_h}$ is the learned parameters. To make $\alpha_i^{k,d}$ comparable over all previous days, it is normalized with a softmax function after the attention scoring

$$\alpha_i^{k,d} = \frac{\exp \left(f_k \left(\mathbf{v}_i^t, \mathbf{h}_i^{k,t-d} \right) \right)}{\sum_{d=0}^{T_o-1} \exp \left(f_k \left(\mathbf{v}_i^t, \mathbf{h}_i^{k,t-d} \right) \right)} \quad (8)$$

where $f_k : \mathbb{R}^{H_v} \times \mathbb{R}^{H_h} \rightarrow \mathbb{R}$ is the scaled time-aware scoring function, defined as follows:

$$f_k \left(\mathbf{v}_i^t, \mathbf{h}_i^{k,t-d} \right) = \frac{\sigma \left(- \left(\mathbf{W}_{fv} \mathbf{v}_i^t \right)^\top \left(\mathbf{W}_{fh}^k \mathbf{h}_i^{k,t-d} \right) \right)}{\sqrt{H_f} \exp(\theta_k d)} \quad (9)$$

where the scaled score $\sqrt{H_f}$ is devoted to normalizing the dot product. Considering that stocks may react differently to different historical interventions, we define a time decay coefficient $\exp(\theta_k d)$ which is sensitive to interventions due to the learned parameter θ_k . Furthermore, the vectors $\mathbf{v}_i^t$ and $\mathbf{h}_i^{k,t-d}$ are transformed into H_f -dimensional vectors using two distinct learned layers $\mathbf{W}_{fv} \in \mathbb{R}^{H_f \times H_v}$ and $\mathbf{W}_{fh}^k \in \mathbb{R}^{H_f \times H_h}$. The key insight here is that $\mathbf{v}_i^t$ has integrated the historical time-series features and we gauge the sufficiency of the intervention through the dot

product (where the similarity reflects whether a stock has been influenced). The presence of $\mathbf{v}_i^t$ as a posterior reference supports this sufficiency test and the dot product serves as a form of posterior probability. The higher the posterior probability, the greater the probability that $\mathbf{h}_i^{k,t-d}$ has already intervened on the target stock, indicating a higher degree of sufficiency in this intervention. Consequently, $\mathbf{h}_i^{k,t}$ would allocate less attention to $\mathbf{h}_i^{k,t-d}$.

c) *Graph-level heterogeneous attention*: To identify the offset embedding $\mathbf{u}_i^{k,t} \in \mathbb{R}^{H_u}$ encompassing all intervention effects, we employ a cross-graph attention mechanism for offset component learning, denoted as follows:

$$\mathbf{u}_i^t = \sigma \left(\sum_{k \in \mathcal{R}} \beta_i^k \mathbf{W}_{uh}^k \tilde{\mathbf{h}}_i^{k,t} \right) \quad (10)$$

where the attention score $\beta_i^k \in \mathbb{R}$ indicates the intensity of intervention k for stock i . It is calculated by an attention layer with softmax normalization

$$\beta_i^k = \frac{\exp \left(g \left(\mathbf{v}_i^t, \tilde{\mathbf{h}}_i^{k,t} \right) \right)}{\sum_{k \in \mathcal{R}} \exp \left(g \left(\mathbf{v}_i^t, \tilde{\mathbf{h}}_i^{k,t} \right) \right)} \quad (11)$$

to make β_i^k comparable over all interventions, where $g : \mathbb{R}^{H_v} \times \mathbb{R}^{H_h} \rightarrow \mathbb{R}$ is the scaled intervention-aware scoring function, defined as follows:

$$g \left(\mathbf{v}_i^t, \tilde{\mathbf{h}}_i^{k,t} \right) = \frac{\sigma \left(\left(\mathbf{W}_{gv} \mathbf{v}_i^t \right)^\top \left(\mathbf{W}_{gh}^k \tilde{\mathbf{h}}_i^{k,t} \right) \right)}{\sqrt{H_g}} \quad (12)$$

where $\mathbf{v}_i^t$ and $\tilde{\mathbf{h}}_i^{k,t}$ are mapped into H_g -dimensional vectors using two different learned layers $\mathbf{W}_{gv} \in \mathbb{R}^{H_g \times H_v}$ and $\mathbf{W}_{gh}^k \in \mathbb{R}^{H_g \times H_h}$. In contrast to (9), there is no posterior reference in this case, and we gauge the necessity of the intervention through the dot product, which serves as a form of prior probability. With a higher prior probability, the intervention becomes more essential, and the offset embedding $\mathbf{u}_i^{k,t}$ directs greater attention toward it.

d) *Discussion in extreme market conditions*: Note that our model is based on modeling with a time granularity of trading days. In extreme market conditions, such as market crashes and Black Swan events, systemic and pervasive volatility may instantaneously propagate throughout the entire financial market, rather than through the relatively moderate lead-lag effect. At this point, the coarse time granularity may be insufficient to deal with such exceptional circumstances, rendering HD²AT potentially ineffective to a certain extent. Therefore, to address these conditions, consideration may need to be given to finer time granularity or the exploration of human-machine hybrid intelligence approaches. However, as demonstrated in the stress test (Section V-F), our model remains capable of mitigating certain risks even in the occurrence of a market crash.

C. Return Prediction With Signal Fusion

To make predictions on stock returns using response embedding and offset embedding, we employ a dense feed-forward layer to seamlessly integrate both representations

$$\hat{y}_i^t = \mathbf{a}_i^\top [\mathbf{v}_i^t \parallel \mathbf{u}_i^t] + b_i \quad (13)$$

where $\hat{y}_i^t$ is the predicted return of stock i on the day t , $\mathbf{a}_i \in \mathbb{R}^{H_v+H_u}$ and $b_i \in \mathbb{R}$ are learned weight vector and bias, respectively. Note that the set of optimized parameters is $\Omega = \{(\mathbf{a}_i, b_i) | i \in \mathcal{V}\}$, with parameter configurations tailored for individual stocks.

Finally, the loss between predictions and targets is back-propagated for optimization, which is defined as follows:

$$\mathcal{L} = \sum_t \sqrt{\frac{1}{N} \sum_{i \in \mathcal{V}} (\hat{y}_i^t - y_i^t)^2} \quad (14)$$

where $\mathcal{L}$ computes the root mean square error (RMSE) between the predicted return $\hat{y}_i^t$ and the ground-truth y_i^t of each stock. By minimizing $\mathcal{L}$, HD²AT learns all parameters to enhance return prediction performance.

V. EXPERIMENTS

In this section, we conduct a series of experiments to evaluate the performance of HD²AT.

A. Evaluation Settings

1) *Datasets*: We collected historical trading data for NASDAQ 100 entities and S&P 500 entities from January 2015 to December 2019 and January 2014 to December 2019, respectively, sourced from Yahoo Finance². In our evaluation, we included 505 stocks from the S&P 500 index and 121 stocks from the national association of securities dealers automated quotation (NASDAQ) 100 index, all of which had complete transaction data throughout the entire period. Each dataset was split into two periods: January 2017 to December 2019 for optimizing the stock return prediction framework (namely response embedding, offset embedding, and signal fusion), and the other for the pretraining of the intervention estimators. For the stock return prediction framework, we adopted a rolling window fashion, where each 25 months of experimental data was split chronologically into 23 months for training, 1 month for validation, and 1 month for testing. The time window scrolls forward for one month in 12 subexperiments.

2) *Compared Baselines*: Our proposed model is compared with a range of methods, encompassing various categories. These include two classical time-series methods, namely GRU [25] and LSTM [24]. Furthermore, we evaluate our model against eight SOTA graph-based methods that have demonstrated success in the domain of stock price prediction, i.e., multipronged attention network for stock forecasting (MAN-SF) [11], spatio temporal hypergraph attention network for stock ranking (STHAN-SR) [34], efficient stock integration with temporal generative filters and wavelet hypergraph attentions (ESTIMATE) [15], THGNN [16], TGC [17], FinGAT [43], ADGAT [5], and DANSMP [13]. Note that, for a fair comparison in the same prediction task, we make slight modifications to the objective functions of these methods to predict future stock returns. Moreover, we collected media news for entities on both datasets spanning from January 2014 to December 2019 from Benzinga³ for those using news and then processed these raw news data following their original procedures, such as sentiment extraction or sentence embedding.

²<https://finance.yahoo.com/>

³<https://github.com/Benzinga/benzinga-python-client>

TABLE II
HYPERPARAMETER SETTINGS OF HD²AT
ON TWO DATASETS

Hyperparameters	NASDAQ 100	S&P 500
Scaling parameter γ	1.8	2.2
Window size T_r	24	24
Window size T_o	12	15
Dimension H_v	128	128
Dimension H_h	96	128
Dimension H_f	128	160
Dimension H_u	96	96
Dimension H_g	64	96
Learning rate	1e-4	5e-5

3) *Evaluation Metrics*: Stock return prediction is studied as a regression task. We utilize two standard metrics, RMSE and mean absolute error (MAE), to measure the deviation between the predicted values and the actual returns. In practice, we can gauge whether a stock rises or falls according to the positive or negative predicted return. Then Mathew's correlation coefficient (MCC) [44] is applied to evaluate the quality of stock movement prediction. On the other hand, a trading portfolio is simulated to evaluate the performance of different models through back-testing, considering the metrics including annualized return rate (ARR), maximal drawdown (MDD), and annualized sharp ratio (ASR). Specifically, ARR is calculated by summing the daily returns, offering a direct measure of annualized profitability averaged over all years. MDD serves as a risk indicator representing the maximum loss of return from peak to trough. ASR assesses returns relative to volatility, providing insights into the risk-adjusted performance.

4) *Implementation Details*: For the intervention estimators, we select the XGBRegressor [40] as the base learner. Furthermore, for the proposed framework, the grid search is employed to determine the optimal hyperparameters to RMSE. All parameters are trained for a maximum of 200 epochs with the Adam optimizer. Table II presents the optimal settings of our model. On average, each training procedure (including pretraining) costs 6.2 and 11.4 h and each test day costs 0.14 and 0.25 s with one NVIDIA Titan V graphics processing unit (GPU) on NASDAQ 100 and S&P 500 datasets, respectively. This implies that our model possesses the capability for fast updates and iterations to dynamically adapt to changes in the economic environment.

5) *Trading Portfolio*: To further evaluate the profitability of our approach, a trading strategy was executed to simulate the investment, i.e., back-testing. To be specific, stocks within the stock pool are ranked based on their return predictions on each trading day, with the highest-ranked stocks at the top and the lowest at the bottom. Then we purchase a maximum of four highest-ranked stocks in the top 15 stocks that are not already in our portfolio and sell an equal quantity of the lowest-ranked stocks. That controls the turnover rate of the strategy and ensures that we consistently hold 15 stocks in our portfolio. We assumed an initial account capital of 1e-7 (10 million) and account for transaction costs of 0.05% when buying shares and 0.15% when selling shares.

TABLE III
PERFORMANCE EVALUATION OF STOCK RETURN PREDICTION (MEAN $\pm$ STANDARD DEVIATION)

Methods	NASDAQ 100			S&P 500		
	RMSE	MAE	MCC	RMSE	MAE	MCC
GRU	2.314e-2 $\pm$ 1.216e-3	1.655e-2 $\pm$ 1.404e-3	0.105 $\pm$ 0.091	2.372e-2 $\pm$ 1.839e-3	1.664e-2 $\pm$ 1.782e-3	0.096 $\pm$ 0.073
LSTM	2.240e-2 $\pm$ 1.585e-3	1.592e-2 $\pm$ 1.743e-3	0.125 $\pm$ 0.077	2.235e-2 $\pm$ 1.972e-3	1.608e-2 $\pm$ 1.532e-3	0.127 $\pm$ 0.089
MAN-SF	1.904e-2 $\pm$ 9.638e-4	1.377e-2 $\pm$ 1.182e-3	0.176 $\pm$ 0.036	1.968e-2 $\pm$ 1.121e-3	1.443e-2 $\pm$ 8.318e-4	0.166 $\pm$ 0.078
STHAN-SR	1.873e-2 $\pm$ 8.275e-4	1.313e-2 $\pm$ 7.452e-4	0.191 $\pm$ 0.027	1.913e-2 $\pm$ 1.231e-3	1.380e-2 $\pm$ 1.125e-3	0.178 $\pm$ 0.061
ESTIMATE	1.913e-2 $\pm$ 8.165e-4	1.376e-2 $\pm$ 9.058e-4	0.182 $\pm$ 0.041	1.961e-2 $\pm$ 1.004e-3	1.466e-2 $\pm$ 1.167e-3	0.165 $\pm$ 0.085
THGNN	1.849e-2 $\pm$ 1.063e-3	1.317e-2 $\pm$ 8.349e-4	0.194 $\pm$ 0.069	1.889e-2 $\pm$ 8.405e-4	1.374e-2 $\pm$ 6.287e-4	0.179 $\pm$ 0.042
TGC	1.816e-2 $\pm$ 6.582e-4	1.281e-2 $\pm$ 5.933e-4	0.205 $\pm$ 0.042	1.934e-2 $\pm$ 7.122e-4	1.412e-2 $\pm$ 9.977e-4	0.171 $\pm$ 0.066
FinGAT	1.636e-2 $\pm$ 1.163e-3	1.153e-2 $\pm$ 1.117e-3	0.229 $\pm$ 0.058	1.887e-2 $\pm$ 6.912e-4	1.321e-2 $\pm$ 8.471e-4	0.184 $\pm$ 0.043
AD-GAT	1.807e-2 $\pm$ 5.401e-4	1.258e-2 $\pm$ 7.984e-4	0.218 $\pm$ 0.055	1.862e-2 $\pm$ 1.085e-3	1.305e-2 $\pm$ 8.348e-4	0.199 $\pm$ 0.076
DANSMP	1.608e-2 $\pm$ 7.592e-4	1.042e-2 $\pm$ 9.938e-4	0.242 $\pm$ 0.043	1.838e-2 $\pm$ 9.856e-4	1.277e-2 $\pm$ 7.714e-4	0.211 $\pm$ 0.069
HD ² AT	1.491e-2 $\pm$ 9.921e-4	1.015e-2 $\pm$ 8.693e-4	0.259 $\pm$ 0.042	1.617e-2 $\pm$ 1.224e-3	1.196e-2 $\pm$ 9.507e-4	0.234 $\pm$ 0.053

Note: Bold values for HD²AT outperforms those of the baselines.

B. Stock Return Prediction

Table III presents the results of stock return prediction on the NASDAQ 100 and S&P 500 datasets, where the metrics are computed across all of the stocks within each prediction period and the means and standard deviations of them are calculated on the 12 prediction periods.

1) *Main Results:* Given the minuscule standard deviations reported in Table III, we leverage the Diebold–Mariano test [45] to examine whether there are statistically significant differences among different models in prediction accuracy across all test days. As shown in Table IV, our model demonstrates superiority over the majority of models at a significance level of at least 10%. For each baseline model, our model exhibits significantly better predictive accuracy on at least one dataset. Overall, these results in both Tables III and IV indicate that our HD²AT consistently outperforms all baseline models for predicting stock returns. The minimum RMSE and MAE scores achieved by HD²AT signify that its stock return predictions closely align with the actual returns, indicating high predictive accuracy. Furthermore, the maximal MCC score reflects the model’s effectiveness in predicting stock movements, specifically whether they will rise or fall. This confirms the effectiveness of our method in capturing structural dynamics through the consideration of mutual interventions and lead-lag dynamics by employing a time-aware heterogeneous attention mechanism.

2) *Analysis:* 1) Both GRU and LSTM, due to their limited ability to comprehend the intricate interdependencies within the stock network, exhibit inferior performance compared to graph-based methods. This illustrates that depending solely on time-series price signals is insufficient to support stock return prediction [12], [13], [28]. 2) MAN-SF, STHAN-SR, and ESTIMATE accounting for stock relations or even heterogeneous relations, outperform time-series models. However, they demonstrate inferior performance over HD²AT due to their limited consideration of the time-varying connections. 3) The stock networks employed by THGNN and TGC are structurally dynamic, leading to a slight improvement in their predictive performance. However, they are surpassed by AD-GAT and HD²AT due to their failure to capture the lead-lag interactions among stocks. 4) FinGAT and DANSMP construct intricate

TABLE IV
DIEBOLD-MARIANO TEST RESULTS FOR
MEAN SQUARED ERROR (MSE)
COMPARISON BETWEEN THE HD²AT
AND BASELINE MODELS

Baselines	NASDAQ 100	S&P 500
GRU	0.0000***	0.0000***
LSTM	0.0000***	0.0000***
MAN-SF	0.0002***	0.0884*
STHAN-SR	0.1127	0.0000***
ESTIMATE	0.0462**	0.0039***
THGNN	0.0925*	0.0196**
TGC	0.0089***	0.0218**
FinGAT	0.0012***	0.1439
AD-GAT	0.0547*	0.0363**
DANSMP	0.0969*	0.0756*

Note: *, **, and *** represent statistical significance at the levels of 10%, 5%, and 1%, respectively.

dynamic heterogeneous graph networks, encompassing inter and intrasector relations, executives relations, and management relations. These methods result in a notable enhancement in prediction performance, particularly in the NASDAQ 100 dataset. Nevertheless, besides these two methods, all graph-based models have ignored the causal interventions arising from daily trading signals of stocks. Consequently, they are all outperformed by HD²AT in terms of RMSE, MAE, and MCC. 5) HD²AT outperforms some methods that leverage media news for prediction, such as AD-GAT, and DANSMP. This observation implies that intervention effects prove to be more effective than news events. Interventions, as outlined in Section II-B2, are triggered by price fluctuations, thus reflecting specific events occurring within the stocks themselves, e.g., dividend changes, and shareholder shifts. This quantified form of intervention is more direct and specific compared to the influence of news events.

C. Back-Testing Profitability

We execute a trading strategy (Section V-A5) in the NASDAQ 100 and S&P 500 datasets to assess the model’s profitability during the test period, and the experimental results are

TABLE V
PERFORMANCE EVALUATION OF BACK-TESTING PROFITABILITY

Methods	NASDAQ 100			S&P 500		
	ARR	MDD	ASR	ARR	MDD	ASR
Benchmark	0.395	-0.111	2.113	0.298	-0.068	2.157
GRU	0.162	-0.068	1.782	0.153	-0.057	1.753
LSTM	0.193	-0.076	1.813	0.191	-0.061	1.889
MAN-SF	0.252	-0.105	1.718	0.240	-0.091	1.652
STHAN-SR	0.286	-0.093	2.046	0.255	-0.083	1.798
ESTIMATE	0.278	-0.088	2.081	0.236	-0.062	2.090
THGNN	0.311	-0.078	2.356	0.269	-0.055	2.482
TGC	0.334	-0.072	2.706	0.248	-0.076	1.964
FinGAT	0.401	-0.063	3.253	0.297	-0.068	2.684
AD-GAT	0.350	-0.072	2.878	0.318	-0.064	2.951
DANSMP	0.416	-0.069	3.342	0.342	-0.042	3.297
HD ² AT	0.437	-0.056	3.478	0.376	-0.039	3.484

Note: Bold values for HD²AT outperforms those of the baselines.

displayed in Table V. To enable comparisons with the broader market trends, we introduced the NASDAQ 100 Index into the NASDAQ 100 dataset and the S&P 500 Index into the S&P 500 dataset, labeling them as “Benchmark” in the table.

1) *Main Results*: The results unequivocally demonstrate that our HD²AT outperforms all other models in the back-testing investment simulation. The highest ARR score underscores its exceptional profitability, the lowest MDD highlights its remarkable risk resilience, and the highest ASR signifies its outstanding overall performance in terms of risk-adjusted returns.

2) *Analysis*: 1) Combining the findings with Table III, we observe that, as a general rule, the more accurate a model’s return predictions are, the stronger its earning potential. This aligns well with investment experience. 2) GRU and LSTM treat each stock as an isolated node, which partially mitigates some external risks, but simultaneously hinders them from modeling momentum spillover, ultimately limiting their profitability. 3) MAN-SF, STHAN-SR, and ESTIMATE cannot capture the real-time dynamics of the network, thus missing out on timely risk feedback and lacking the ability to make real-time risk adjustments. As a result, their ASR performance lags behind that of even the benchmarks. 4) TGC and AD-GAT overlook the heterogeneity of the stock network, while THGNN, FinGAT and DANSMP disregard the lead-lag dynamics, which leads to their worse performance compared to our HD²AT. Furthermore, HD²AT models interventions through causal inference, making it more robust and sensitive to risks, thus outperforming all baselines across all metrics.

D. Ablation Study

To systematically verify the efficacy of each component within our model, we design seven variants of HD²AT: 1) *w/ correlations*, which computes the edge weights using partial correlation coefficients of different indicators. 2) *w/o dynamics*, which employs intervention effects averaged from the previous year of the prediction day. 3) *w/ explicit relations* (*w/ explicit rels.*), which replaces the interventions with static industry category, supply chain, business partnership, and investment relations [13], [17]. 4) *w/o dynamic attention*

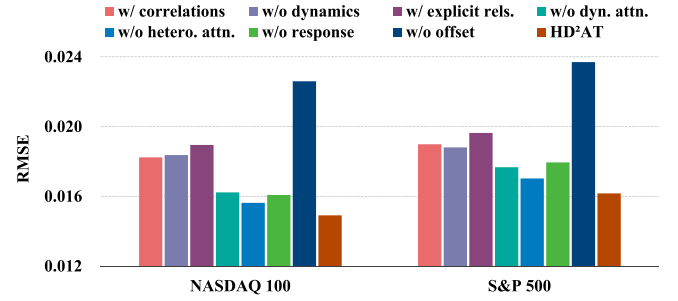


Fig. 4. RMSE performance of different variants on two datasets.

(*w/o dyn. attn.*), which removes the time-level dynamic attention and exclusively focuses on the last intervention embedding. 5) *w/o heterogeneous attention* (*w/o hetero. attn.*), which removes the graph-level heterogeneous attention and averages all heterogeneous intervention embeddings. 6) *w/o response*, which omits response embeddings in the signal fusion module. 7) *w/o offset*, which omits offset embeddings in the signal fusion module. Note that *w/o dynamics* and *w/ explicit relations* need to eliminate dynamic attention like *w/o dynamic attention*, as it is unnecessary for these static stock networks.

1) *Main Results*: The RMSE performance, as depicted in Fig. 4, highlights the conclusive role of different components within our HD²AT model. The inclusion or exclusion of specific components within HD²AT has a significant impact on its prediction performance.

2) *Analysis*: 1) *w/o dynamics* and *w/ explicit relations*, which lack structural dynamics or causal interventions, struggle to effectively capture real-time interactions and thus fall short when compared to even *w/o dynamic attention*. 2) When we compare variants lacking the dynamic network inference module (specifically, *w/ correlations*, *w/o dynamics*, *w/ explicit rels.*, and *w/o offset*) to those that retain this module (specifically, *w/o dyn. attn.*, *w/o hetero. attn.*, and *w/o response*), the underwhelming performance of the former underscores the effectiveness of incorporating heterogeneity, structural dynamics, and causal interplay within the stock network. This dynamic network inference module makes the most significant contributions to predictive performance and, as a result, forms the cornerstone for subsequent modules. 3) Failing to incorporate dynamic attention or heterogeneous attention leads to inadequate exploitation of spatial and temporal information present in the intervention graphs. Owing to this, *w/o dyn. attn.* and *w/o hetero. attn.* exhibit subpar performance. 4) In the absence of offset embeddings (*w/o offset*), HD²AT essentially degrades into an LSTM, yielding similar predictive performance of LSTM as indicated in Table III. 5) *w/o response* nearly outperforms all SOTA baselines. This can be attributed to the fact that even when response embedding is removed in the fusion module, the response information is implicitly present within the intervention embedding as illustrated in Fig. 2. The design of node-level propagation compels the model to revise its understanding of the offset, thereby merging the offset embedding with some aspects of the response information. However, this merging may lead to signal overlap within the

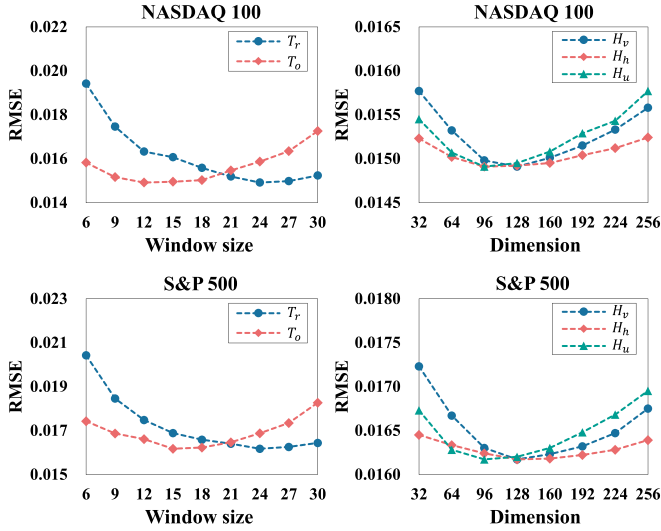


Fig. 5. Parameter sensitivity analysis w.r.t. the lookback window size and the dimension of representations.

offset embedding, resulting in a lower RMSE score than HD^2AT . This highlights the importance of considering both response and offset as integral components of the model.

E. Parameter Sensitivity Analysis

To comprehend the performance of HD^2AT under different hyperparameter settings, we examine its parameter sensitivity. Fig. 5 illustrates how the RMSE score is influenced by alterations in window sizes (i.e., T_r and T_o), as well as changes in dimensions (i.e., H_v , H_h , and H_u).

1) *Window Sizes T_r and T_o* : We analyze the performance variation with different window sizes in the left column of Fig. 5. We can find that reducing the value of T_r causes the response embedding to overlook long-term trends, leading to a steep increase in the RMSE score, and larger T_r may deteriorate performance by introducing stale information from past days. Similar trends are observed for T_o , but HD^2AT shows less sensitivity to a decrease in T_o and more sensitivity to an increase, attributed to the temporal decay mechanism in the time-level dynamic attention module. Lower T_o essentially directs HD^2AT to focus on recent interventions, while higher values introduce more noise, both diminishing HD^2AT 's performance in different degrees.

2) *Dimensions H_v , H_h , and H_u* : The right column of Fig. 5 suggests that excessively low or high embedding dimensionality leads to a reduction in prediction quality due to underfitting and overfitting, respectively. We note that HD^2AT is less sensitive to H_h than to H_v and H_u . This is because the intervention embedding, with a dimension of H_h , is located in the hidden layer during prediction. Furthermore, the model's varying degrees of dependence on response embedding (with a dimension of H_v) and offset embedding (with a dimension of H_u) result in differing effects on predictions when these dimensions are adjusted.

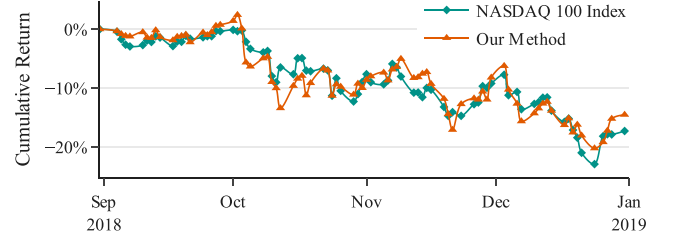


Fig. 6. Stress test during the market crash period.

F. Robustness to Market Crash

To verify the robustness of our model against extreme market conditions, we conduct a stress test during the market crash period from October 2018 to December 2018. Due to the Federal Reserve's persistent interest rate hikes in the first half of 2018, the stock market came under pressure, ultimately experiencing a widespread and highly volatile downturn. To vividly illustrate the market's rapid decline, we tracked the NASDAQ 100 Index and the cumulative returns of our method from the stable September in Fig. 6. As the index encountered successive sharp declines from October, HD^2AT struggled to navigate the market's panic sentiment and even exhibited instances of erroneous judgments. However, as it progressively acclimated to the market conditions within the time window, the model showcased an aptitude for risk management to a certain degree, avoiding more significant economic losses. This indicates that our model demonstrates a level of resilience in the face of extreme market conditions but falls short of generating profits. In practical investment scenarios, there is often a greater emphasis on risk management capabilities than on profit-generating capabilities.

G. Case Study on Interpretability

To further investigate how HD^2AT captures spatial structural dynamics and temporal lead-lag dynamics, we conducted a case study on the NASDAQ 100 test set to explore its interpretability. On one hand, we focus on the specific intervention graph concerning "high trading volume with significant positive return," whose average node degree varies with overall market fluctuations, as depicted in the bottom of Fig. 7. When the market sentiment is bullish and the Index exhibits a significant upward trend, our intervention estimator effectively identifies a surge in the number of such interventions (with significant edge weights). Conversely, during bearish market phases, this figure diminishes. This observation underscores the dynamic information encapsulated within the network structure inferred by these estimators. On the other hand, we average all the attention scores of all individual stocks on a given day, as exemplified in the top of Fig. 7. we can discern a pattern wherein HD^2AT effectively allocates attention to the days with sustained interventions to subsequent trading days. Notably, during the positive market performance period (April to May), the temporal dynamic attention does not particularly emphasize this timeframe, due to the Index's lack of significantly high returns. It is possible that other interventions, such as

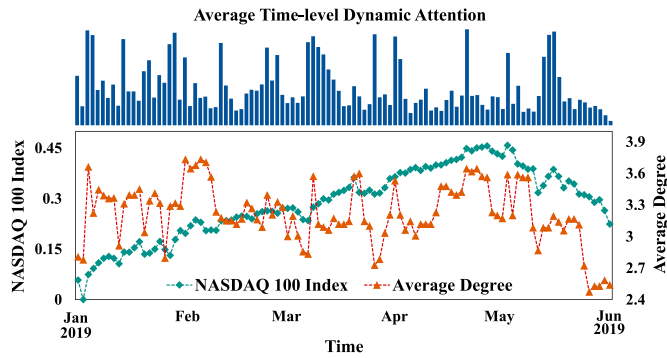


Fig. 7. Visualization of the spatial-temporal dynamics. *Bottom*: The relative movement of NASDAQ 100 Index and the time-varying average degree of the intervention graph of “high trading volume with significant positive return.” *Top*: The evolving average time-level dynamic attention.

continuous uptrends, would better capture this period. This further illustrates how different types of interventions serve different analytical purposes.

VI. CONCLUSION AND FUTURE WORK

In this article, we propose the HD²AT for stock return prediction. It comprehensively models the intrinsic stock interactions across four key dimensions: heterogeneity, causality, structural dynamics, and lead-lag dynamics. First, by leveraging the intervention estimators, HD²AT dynamically infers the heterogeneous stock network. Second, our model generates the response embeddings to quantify how each stock responds to its historical signals. Third, based on the constructed intervention graphs, the offset embedding is determined through node-level intervention propagation, time-level dynamic attention, and graph-level heterogeneous attention, to encapsulate the extent of interventions from neighboring stocks. Finally, a signal fusion module is designed to seamlessly integrate both representations for prediction. The experimental results of stock return prediction demonstrate that HD²AT achieves the minimal average errors (RMSE and MAE) and maximum prediction accuracy (MCC) across two stock datasets with different numbers of nodes. Moreover, it attains the highest annualized return (ARR) while controlling for risk (ASR) and minimizing maximum drawdown (MDD) when back-testing. The proposed method holds the potential to extend to other time-varying graph learning domains like topology tracking for power systems and latent social network inference, whose full power remains to be harnessed.

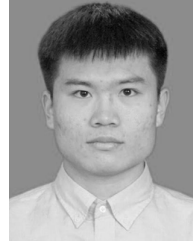
The future extension of our HD²AT framework is to explore intervention networks incorporating different financial instruments or even web media. In this context, the modeling of the comprehensive intervention network requires consideration of not only edge heterogeneity but also node heterogeneity, encompassing various node types such as options contracts, underlying assets, and news. Simultaneously, quantifying causality between nontime-series and time-series data, e.g., between news text and price signals, poses a crucial challenge. We believe that under ethical and regulatory oversight, the inclusion

of more market information and external news will significantly enhance the accuracy of return predictions.

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Haotian Liu received the B.S. degree in automation science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2022. He is currently working toward the M.S. degree with the same institution.

His research interests include data mining, graph learning, and relevant applications in finance.



Yadong Zhou (Member, IEEE) received the B.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 2004 and 2011, respectively.

He was a Postdoctoral Researcher with the Chinese University of Hong Kong, Hong Kong, from 2014 to 2015. He is currently a Professor with the Faculty of Electronic and Information Engineering, Xi'an Jiaotong University. His research interests include data analysis and mining, network science and its applications, and software defined networks.

He has published more than 60 papers in prestigious refereed conferences and journals, such as IEEE TRANSACTIONS ON SMART GRID, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, Association for Computing Machinery Transactions on Knowledge Discovery from Data (ACM TKDD), IEEE International Conference on Computer Communications (INFOCOM), and Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence (AAAI).



Yuxun Zhou received the B.S. degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2009, the diplôme d'Ingenieur degree in applied mathematics from Ecole Centrale Paris, Gif-sur-Yvette, France, in 2012, and the Ph.D. degree in electrical engineering from the Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, CA, USA, in 2017.

His research interests include artificial intelligence, statistical learning theory, and paradigms for modern information-rich, large-scale, and human-

involved systems.



Bowen Hu (Graduate Student Member, IEEE) received the B.S. degree in automation science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2019. He is currently working toward the Ph.D. degree with the same institution.

His research interests include data analysis and mining.